

Statistical Testing: Chi-Square Test of Independence

To determine whether categorical variables are significantly associated with booking cancellations, a Chi-Square Test of Independence was performed.

Hypotheses

Null Hypothesis (H_0): There is no association between the selected categorical variable and booking cancellation status.

Alternative Hypothesis (H_1): There is a significant association between the selected categorical variable and booking cancellation status.

Example: Market Segment vs Cancellation

A contingency table was created between Market Segment and Booking Cancellation Status, followed by the Chi-Square test.

Results Interpretation

If the p-value is less than 0.05:

- Reject the null hypothesis.
- Conclude that Market Segment significantly influences booking cancellations.
- The relationship observed is unlikely to have occurred by random chance.

If the p-value is greater than 0.05:

- Fail to reject the null hypothesis.
- No statistically significant relationship exists between the variables.

Business Implication

The results indicate that customer acquisition channels and market segments play an important role in cancellation behavior. Understanding these patterns enables hotel managers to design targeted retention strategies and optimize pricing policies for high-risk segments.